Autonomous Underwater Vehicle

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About Us

Cyclone RoboSub, founded in 2023, is an engineering student design team at UC Davis developing an autonomous underwater vehicle (AUV). Our interdisciplinary team of 50+ students is organized into three divisions and six sub-teams, each tackling different aspects of this project.



Figure 1: 2026 Team Photo

RoboSub Competition

Teams from around the world compete with custom AUVs which complete tasks such as manipulating objects, maneuvering around obstacles, and firing torpedoes. Our team prioritized navigation-based tasks, eliminating the need for precise manipulation. Cyclone RoboSub will be competing for the second time this coming July!

Vehicle Design

Our chosen thruster configuration allows for six degrees of freedom while its symmetry simplifies vehicle control and maneuverability. Designed to be slightly positively buoyant and have a low center of mass, the vehicle naturally corrects errors in orientation. All electronics are housed in a 6" acrylic tube with Blue Robotics WetLink penetrators to connect cables to the external environment. Forward and downward facing cameras are enclosed in custom SLA printed watertight enclosures.

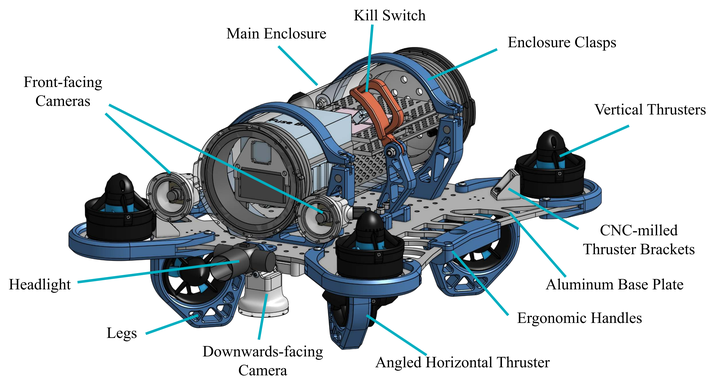


Figure 3: Structural CAD of Robot (Made in Onshape)

Vision

Two cameras provide video feeds, which are processed by a YOLO vision model to identify objects within the pool and assign position estimates. These estimates are used to inform vehicle navigation and are constantly updated.

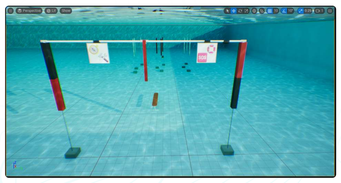


Figure 4: Unreal simulation of competition

Vision data can be simulated in Unreal Engine and passed into MATLAB Simulink to seamlessly test vision integration.

Sensors and Controls

The vehicle relies on a combination of sensors to determine its depth, heading, and position within the pool. A Waterlinked A50 DVL employs a 4-beam sonar system to determine the vehicle's velocity relative to the pool bottom. A sensor fusion algorithm combines velocity data from the DVL and acceleration data from an onboard InertialSense IMU to estimate the vehicle's position.

All sensor data and vehicle commands are sent over ROS2 which integrates directly with a C-code generated controller. The controller, built in MATLAB Simulink, uses a 6-axis, cascaded PID feedback system to navigate the vehicle to desired waypoints. The controller also has the ability to incorporate position estimates from the stereo vision system to refine its position estimates.

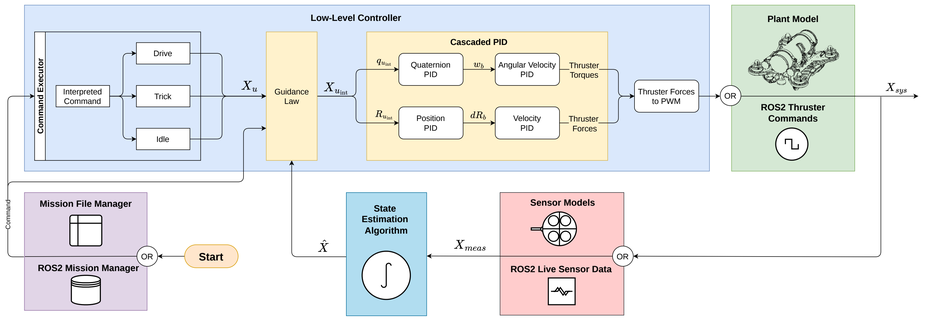


Figure 5: Simulink Controller Diagram

Software Architecture

The robot's software architecture is broken up in a way that allows the vehicle to operate in different modes depending on the use case and test setup. For manual control, PS5 controller inputs are mapped to basic vehicle maneuvers. For testing, the software architecture allows for a hardware-in-the-loop setup where a laptop, running MATLAB, can be inserted in place of a C-code generated controller, allowing for on-the-fly controller tuning and adjustments. The system has the ability to interpret both simulated sensor data or live senor data, minimizing the need to actively run the vehicle in the water.

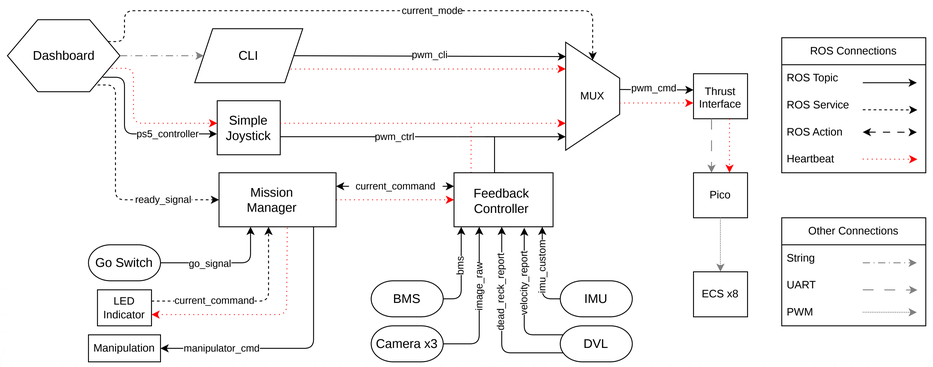


Figure 6: Software System Architecture

For competition runs, the vehicle parses a mission file which defines the maneuvers, actions, and waypoints the vehicle must complete. These goals are interpreted and the controller maneuvers the vehicle accordingly. This is done using a decision tree which incorporates fallbacks in the event of a failed objective, allowing the vehicle to progress and recover on the fly. LED strips indicate the vehicle's state and a heartbeat system allows the vehicle to abort or recover in the event of system interruptions.

Electrical Systems

The vehicle is powered by a 4s Lipo battery and features custom PCBs for battery health, telemetry, and thruster control. A fusebox provides over-current protection and the battery management PCB monitors power draw. System architecture and control software run on a Raspberry Pi 5 while a Jetson Orin Nano handles vison processing.

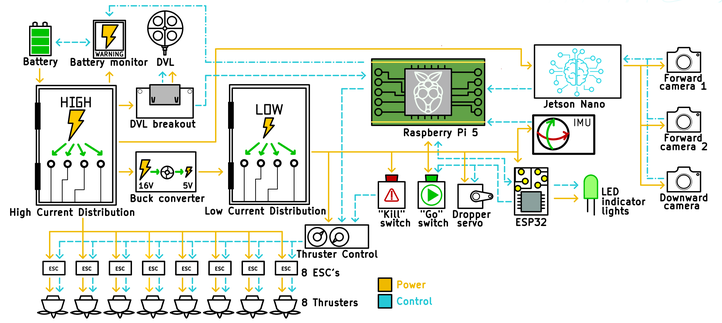


Figure 7: Complete Electrical Diagram

Research

Beyond the competition, Cyclone RoboSub is contributing to environmental research efforts through field deployments and interdepartmental collaborations. To enable parallel research effort progress, the team designed an independent senor module to attach to the underside of the vehicle which measures pH, depth, and temperature, allowing us to collect profiles of the UC Davis arboretum.



Figure 8: Sensor Module

Most recently, Cyclone RoboSub partnered with the UC Davis Tahoe Environmental Research Center to support periphyton imaging efforts in North Lake Tahoe.

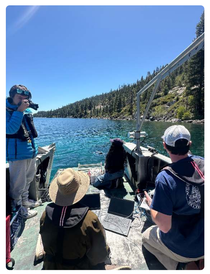


Figure 9: Incline Village Deployment

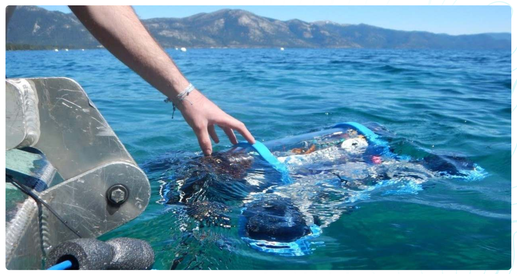


Figure 10: Submerged Robot at Lake Tahoe

Acknowledgements

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